Ergonomic Risk Detection and Mitigation Using YOLOv8 with Vision-Language Models and Retrieval-Augmented Generation

Abstract

Work-related injuries remains a significant and ongoing concern in the construction industry. This research presents a computer-vision based approach for detecting and assessing ergonomic risks among construction workers. The system performs five key functions: (1) assessment of workers' postures; (2) identification of normal and risky physical activities; (3) measurement of activity duration; (4) quantification of high-risk movement frequency; (5) recommendations for risk mitigation using Large Language Models. To validate its performance, the system was tested on video datasets capturing various construction tasks such as drywall finishing, hand screeding, kneeling finishing, laying concrete blocks, and tying rebar. Results were benchmarked against expert annotated groud truth data, which demonstrated high accuracy in detecting ergonomic risks. Overall accuracy exceeded 90% across multiple task categories, which confirms the reliability of the system in detecting both postures and activity transitions. The proposed approach allows construction managers to identify unsafe practices on-site.

Keywords: Ergonomic Risk, Construction Worker, Computer Vision, Vision Language Model, Retrieval-Augmented Generation

1 Introduction

It is reported that the prevalence of work-related injuries among construction workers reached 34% in the US in 2020 [?]. These risks in construction arise from physical, environmental, and organizational factors. Physical demands such as heavy lifting, repetitive movements, prolonged static postures, and awkward positions increase injuries like low back pain and joint disorders [?]. Environmental conditions, including slippery surfaces, poor lighting, extreme temperatures, and disorganized worksites, force unsafe positions and raise physical strain [?]. Organizational issues such as insufficient training, lack of rest breaks, heavy workloads, and poor supervision further amplify risks by pushing workers beyond their physical limits [?]. Given the complex and multifactorial nature of ergonomic risks, efficient monitoring and real-time feedback systems are urgently needed.

Computer vision methods are increasingly being used to automate ergonomic risk detection by analyzing workers' postures and movements through cameras and algorithms [?]. A typical workflow involves applying deep learning and computer vision models to identify key body parts, such as the head, knees, and shoulders, in images captured on construction sites. These identified points are used to construct a skeletal representation of the worker. From this skeletal structure, joint positions are determined, which allows for the calculation of joint angles. These joint angles are then used to assess ergonomic risk using established metrics such as the Ovako Working Posture Analysis System (OWAS) [?], Rapid Upper Limb Assessment (RULA) [?], and Rapid Entire Body Assessment (REBA) [?]. These assessment tools classify specific worker postures into different levels of injury risk.

However, current computer vision and posture estimation methods for ergonomic risk detection face several limitations in real-world construction environments. These include ~~: (1) Predominant focus on posture, while often neglecting other critical ergonomic risk~~ a predominant focus on single-frame posture recognition while neglecting temporal factors such as repetition and ~~the durationof postures; (2) Lack of feedback or corrective measures based on the identifiedergonomic risks~~duration, as well as a lack of mechanisms for delivering corrective feedback after a risk is identified. To address these issues, this research developed anergonomic risk detection system ~~to analyze construction workers' postures and provide~~ that analyzes construction workers' postures and provides actionable feedback to help mitigate musculoskeletal disorders. The ~~key~~ main contributions of this study are ~~: (1) extracting workers' postures, along with the frequency and duration~~

~~of associated ergonomic risks, from video footage captured on construction sites; and (2) utilizing a Large Language Model (LLM)-based~~ summarized as follows:

1. A vision-based pipeline using the *yolov8n-pose* model to detect worker keypoints and estimate posture from construction site videos.
2. A temporal analysis module that tracks bending duration, bending transitions, and trunk-flexion-based REBA scoring across consecutive frames rather than relying only on isolated images.
3. A full workflow for identifying ergonomically risky activities in multiple construction tasks using automated pose estimation, tracking, and activity segmentation.
4. A vision-language-model-assisted component that identifies handled tools or objects and supports approximate REBA load-category interpretation, together with a retrieval-augmented ~~generation (RAG) system to deliver tailored recommendations for reducing injury risk based on the video analytics results.~~ recommendation module for context-aware ergonomic risk mitigation guidance.

To improve the overall clarity of the workflow, Figure 1 provides a graphical abstract of the complete system presented in this paper. It summarizes how construction video frames are processed through pose estimation and tracking, full-body REBA scoring, temporal activity analysis, and the VLM–RAG recommendation module to produce ergonomic risk indicators and mitigation guidance.

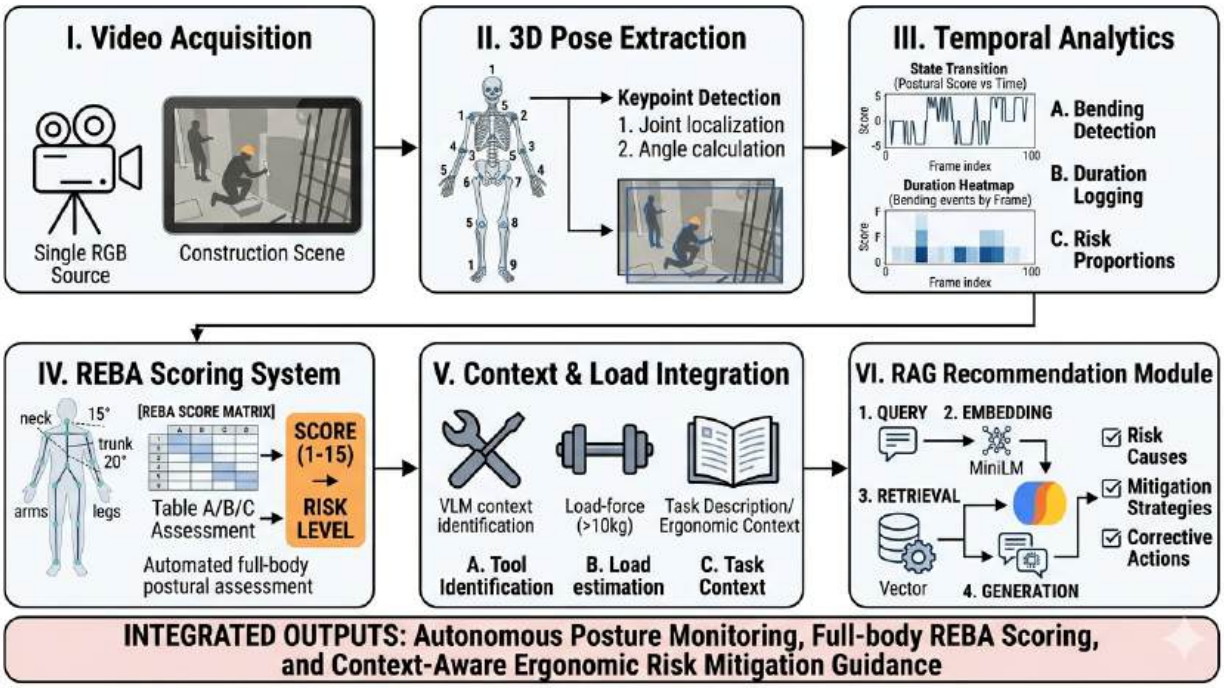


Figure 1: Graphical abstract of the proposed framework. Construction site video is processed through YOLOv8 pose detection, temporal tracking, full-body REBA computation, VLM-assisted approximate load interpretation, and a retrieval-augmented recommendation module to generate ergonomic risk scores, temporal exposure measures, and mitigation guidance.

The remainder of this paper is organized as follows: Section 2 presents a comprehensive literature review covering ergonomic risks in construction, existing assessment methods, and computer vision applications. Section 3 details the proposed methodology, including system architecture, pose estimation techniques, and risk detection algorithms. Section 4 describes the case study implementation and validation results across various construction tasks. Section 5 discuses the findings, limitations, and implications of the research.

2 Literature Review

2.1 ~~Overview of ergonomic risk in construction tasks~~

Construction work is ~~characterized by its physically demanding nature, which often requires workers to engage in repetitive tasks and awkward posturesleading to work-related strain risk [?]. For example, ?] reported that musculoskeletal disorders account for more than 50% of all occupational diseases in the European Union, while in the United States they represent over 30% of non-fatal occupational injuries. Beyond these national statistics, prevalence studies show the extent of~~

~~the problem among construction workers themselves. [?] found that the prevalence of~~ physically demanding and frequently requires repetitive motions, awkward postures, kneeling, stooping, and material handling under dynamic site conditions [?]. As a result, musculoskeletal disorders remain a persistent concern in the industry. Prior studies have reported high prevalence rates of musculoskeletal symptoms among construction workers ~~reached 34% in the US, 41% in Hong Kong, 36% in Ethiopia, and 76% in Malaysia.~~

~~To mitigate the risks associated with musculoskeletal disorders, several strategies have been suggested, including Ergonomic~~ across multiple countries, which highlights the need for scalable and objective ergonomic monitoring methods [? ?]. Common mitigation strategies include ergonomic job design, ~~implementing~~ engineering controls, ~~work practice~~ work-practice controls, administrative ~~controls~~interventions, and early identification of ~~risk factors. These strategies can involve manual observation, the use of wearable sensors, or vision-based tracking systems to monitor workers'~~ hazardous postures and movements ~~and identify potential ergonomic hazards~~ [?].

2.1 ~~Existing methods for ergonomic risk assessment~~

~~Ergonomic assessment methods are crucial for identifying and mitigating risks of musculoskeletal disorders in various industries [?]. These methods can be broadly categorized into three main groups:~~ Ergonomic assessment approaches can generally be grouped into self-report~~methods~~, observation-based~~methods, and direct measurement methods .~~

, and direct-measurement methods [?]. Self-report ~~methods involve gathering data through questionnaires, interviews, or worker diaries to capture workers' perceptions of physical strain, discomfort, and related symptoms. These approaches do not directly measure ergonomic risk factors such as posture or load but instead provide indirect evidence by documenting musculoskeletal problems and perceived exertion. Common tools include~~ tools such as the Nordic Musculoskeletal Questionnaire ~~[?], which records musculoskeletal symptoms,~~ and the Borg ~~Scale [?], which measures perceived exertion.~~

scale capture workers' perceived discomfort and exertion, but they do not directly measure posture or external exposure [? ?]. Observation-based methods ~~rely on systematic evaluation of workers' postures and movements using standardized ergonomic tools such as~~ including OWAS, RULA, and REBA ~~[?]. Traditionally, these assessments were carried out through manual observation by trained experts, but such approaches are prone to subjectivity, human error, and variability in judgment [?]. Recent advances in computer vision have enabled the automation of posture~~

~~recognition and ergonomic scoring, addressing many of these limitations. For example, ?] developed a vision-based system using OpenPose to automatically compute RULA scores, achieving performance comparable to expert assessments even under poor lighting, occlusions, and multi-worker conditions. Similarly, ?] applied the VideoPose3D model combined with a support vector machine classifier to assess lower-body postures from video input, reaching nearly 95% accuracy and providing real-time visual feedback via a dashboard. In another study, ?] introduced a data-driven model with Heuristic Gaussian Cloud Transformation and fuzzy inference to analyze 3D pose estimations from RGB cameras, which significantly improved accuracy and sensitivity over conventional methods. These vision-based approaches reduce time and labor costs, enhance scalability, and provide consistent results across diverse work environments.~~

~~Direct measurement methods involve using sensors and instruments to collect objective data about workers' physical exposures. Techniques include motion capturesystems~~provide structured frameworks for rating posture-related risk, yet manual scoring is time consuming and subject to human variability [? ? ? ?]. Direct-measurement approaches using motion capture, inertial measurement units, pressure insoles, and electromyography ~~to measure joint angles, body movements, and muscle activity. For example, ?] used surface electromyography sensors in combination with machine learning models to directly assess ergonomic risks. Similarly, ?] used wearable Inertial Measurement Units and a machine learning-based data-driven model for direct ergonomic risk assessment of musculoskeletal disorders in construction workers. These methods provide quantitative data continuously which offers high accuracy and detailed insights. However, they can be expensive, complex to implement~~can provide continuous quantitative data, but they often require specialized hardware, increase deployment cost, and may interfere with ~~workers' activities [?]~~worker activity [? ? ?].

2.1 ~~Applications of computer vision for ergonomic assessment~~

~~Computer vision methods for ergonomic assessment have advanced from 2D image-based analysis to 3D pose estimation and automated risk prediction. Early approaches~~ Computer vision research has attempted to address these limitations by automating posture recognition from images and video. Earlier methods relied on 2D ~~cameras with handcrafted features and traditional classifiers . For example, ?] applied view-invariant feature extraction and bagged decision trees to recognize back, arm, and leg postures, achieving up to 97.6% accuracy in dynamic outdoor settings. ?] used body silhouette features and a Support Vector Machine trained on virtual human postures, reaching~~

~~88.6% accuracy. Building on these methods, deep learningenabled~~ image features and conventional classifiers [? ?]. More recent approaches use deep learning, 3D pose estimation~~and integration with ergonomic assessment tools. ?] combined a deep learning-based~~ , and automated ergonomic scoring. For example, ?] integrated 3D ~~model with REBAfor joint-level assessment from 2D videos, reporting 70–96% accuracy in both lab and field experiments. ?] used convolutional neural networks to compute RULA scores directly from images and videos~~ pose estimation with REBA, ?] computed RULA scores from image data under real working conditions~~. Hybrid learning-based methods further extended accuracy and predictive capabilities. For example, ?] employed a convolutional pose machine with a probabilistic neural network, yielding 87.0% accuracy in joint recognition and over 96% accuracy in predicting musculoskeletal disorder risk levels. ?] introduced a data-driven approach using Heuristic Gaussian Cloud Transformation with fuzzy inference and~~ , and ?] combined 3D ~~pose estimation from RGB videos. Recent advances incorporate temporal and motion-based modeling . ?] applied an incremental Convolutional LSTM to posture recognition from wearable IMUs. ?] developed a 3D vision framework integrated with predetermined motion time systems for modular construction tasks. ?] proposed a~~ RGB-based pose estimation with a data-driven inference framework. Temporal modeling has also begun to appear in the literature through approaches such as recurrent models for posture recognition and transformer-based ~~model with self-similarity matrices to automatically count repetitive worker actions.~~

2.1 ~~Research Gap~~

~~Existing approaches to ergonomic risk detection in construction environments often struggle to capture the duration and frequency of risky postures, particularly trunk bending, which are critical indicators of long-term injury risk. Most studies emphasize posture classification at isolated time points and overlook the cumulative effects of repetitive or sustained movements. In addition, current methods rarely provide contextual or adaptive feedback, which limits their usefulness in dynamic construction settings. To overcome these challenges, this research introduces a real-time ergonomic risk detection system with two key innovations: (1) accurate measurementof bending duration and frequency that enables more comprehensive and time-aware ergonomic assessments; and (2) integration of a Retrieval-Augmented Generation (RAG) framework that delivers dynamic, context-aware feedback and recommendations based on detected risks~~ repetitive action counting [? ? ?]. However, most prior studies still emphasize posture recognition or scoring at the frame level. In contrast, the present work combines pose estimation with temporal bending-frequency

tracking, posture-duration measurement, ~~trunk-flexion-based REBA analysis,~~ full-body REBA scoring, VLM-assisted approximate load estimation, and retrieval-augmented LLM feedback so that detected risks can be interpreted and acted upon within a single workflow.

3 Methodology

To enable real-time, low-cost ergonomic risk monitoring in dynamic construction environments, we design a lightweight system deployable with a single RGB camera and an edge device. It comprises a camera that captures video feeds of workers during routine activities. The stream is processed in real time by ~~a YOLOv8 pose estimator~~ the *yolov8n-pose* model to identify each worker and estimate their pose. Capture, processing, and visualization are implemented with OpenCV. The architecture includes modules for pose estimation, activity monitoring, and risk analysis. Figure 2 presents the system architecture for human pose detection.

3.1 YOLOv8 Pose Estimation Pipeline

The ergonomic risk detection system is designed ~~using the YOLOv8 architecture [?]. The system begins with an input image~~ around the YOLO family of one-stage detectors. We cite both the original YOLO formulation by **?**] and the later architectural review by **?**]; the specific pose model used in this study is *yolov8n-pose*. In this paper, the terms "backbone," "neck," and "head" refer to standard stages of the detection network architecture rather than anatomical regions of the worker. The backbone extracts image features, the neck fuses features across scales, and the head predicts person bounding boxes and body keypoints.

The input image is resized to 640×640×3 and ~~processes it through three primary stages: backbone, neck, and head. The backboneextracts key features using~~ processed through these three stages. In the backbone, CBS modules (convolution, batch normalization, and SiLU ~~, the Sigmoid Linear Unit activation) to enhance essential patterns such as edges, shapes, and textures. To capture a wide range of details,~~ activation) extract low- and mid-level visual features. C2f blocks~~(a lightweight cross stage partial design that improves gradient flow and feature reuse) are applied. Stride 2 convolutions reduce the spatial size of the feature maps, and an SPPF block (Spatial Pyramid Pooling Fast) at~~ , which are derived from cross-stage-partial feature reuse principles, improve gradient flow and computation efficiency [?]. An SPPF block near the end of the backbone aggregates multiscale ~~context. Upsampling is performed later in the neck for multiscale feature fusion . The neck serves as the feature aggregation stage, where extracted features are refined and~~

~~prepared for detection. It uses~~ spatial context [?]. The neck performs multiscale fusion using a feature pyramid ~~with~~ and path aggregation, commonly ~~implemented as an FPN (Feature Pyramid Network) with PAN (Path Aggregation Network), to combine information from multiple scales. This enables the system to focus on both small and large details simultaneously. The neck also employs upsampling and concatenation operations, which preserve key details and create a clear and comprehensive feature map for further analysis. The head of the architecture is designed to detect body keypoints and poses. It includes three detection heads, each operating at a different spatial scale for small, medium, and large instances. Each head is anchor free and decoupled, producing a person bounding box, class scores, and K keypoints per instance, where K is the number of defined body keypoints. The higher resolution scale helps localize small structures such as hands and feet, the lower resolution scales capture larger structures such as the torso and hips, and the medium scale supports intermediate parts such as elbows and knees. A minimum confidence threshold of~~ described through the FPN and PAN structures [? ?]. Finally, the decoupled anchor-free head predicts one person bounding box and a set of body keypoints for each detected instance.

A minimum keypoint confidence threshold of 0.5 ~~was selected after experimenting with various values to balance detection accuracy and reliability.~~ was selected after empirical tuning to balance false positives and missed detections. Keypoints whose confidence falls below this threshold are treated as undetected for that frame and are not passed directly to the downstream posture-analysis module. YOLOv8 was selected for the present study because it provides person detection and pose estimation within a single lightweight pipeline that is suitable for near-real-time deployment on ordinary hardware. More specialized pose-estimation toolkits and video-based methods, including RTMPose, MMPose, and VideoPose3D-style temporal reconstruction pipelines, are valuable comparison targets for future work. However, the current benchmark focused on comparing the proposed YOLOv8 variants against an established baseline implementation of OpenPose under the same construction clips and downstream REBA workflow. Additionally, smoothing techniques are used to stabilize the detected keypoints across frames, which prevents jitter in real-time applications.

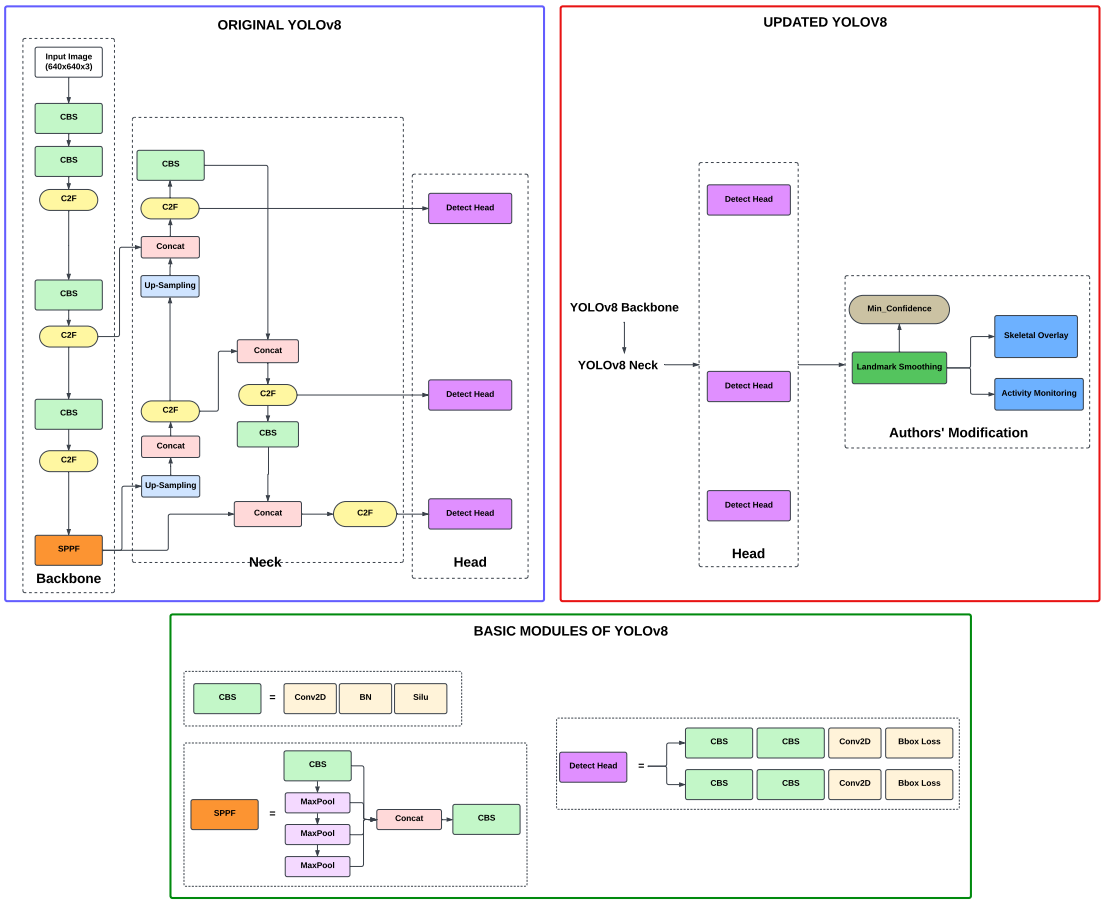


Figure 2: Proposed Architecture

3.2 Pose Estimation and Landmark Smoothing

Pose estimation in this approach relies on ~~YOLOv8~~ *yolov8n-pose* to estimate body keypoints for worker posture assessment: head, shoulders, hips, knees, and ankles. Figure 3 represents the various keypoints of the human body that are identified by the YOLOv8 model. The model identifies these keypoints and makes connections between various poses to create a skeleton of the person detected.

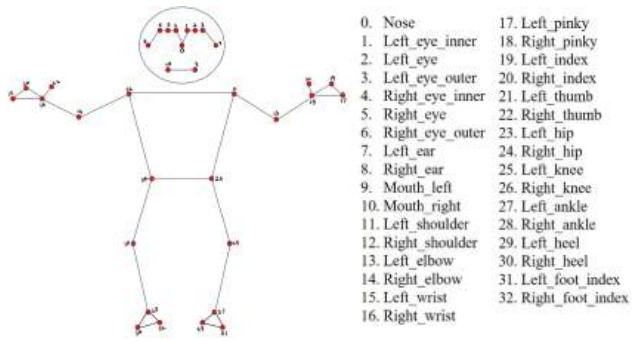


Figure 3: Keypoints identified by YOLOv8 model

A landmark smoothing technique is performed to increase the stability of the predicted keypoints. The moving average is computed over successive frames for each keypoint ~~by the smoothing function~~ to reduce jitter ~~. This ensures that the detected postures reflect the real movements of workers.~~ and suppress short-lived missing detections. In practice, when a keypoint confidence falls below the minimum threshold, the corresponding output coordinate may collapse to the image origin $(0, 0)$. For that reason, the smoothing function acts not only as a jitter filter but also as a lightweight interpolation mechanism that prevents these origin artifacts from propagating directly into downstream posture calculations. A simple moving average was selected over more complex filtering methods due to its computational efficiency and ease of implementation, which are critical for real-time applications. The landmark smoothing scheme significantly improves the coherence of posture detections, particularly in challenging conditions where rapid movements may introduce noise in the detection process. ~~Enhancements were made to the existing YOLO model by integrating a novel smoothing technique which contributed to increasing the~~ The smoothing function averages each keypoint position across successive frames to improve stability and coherence ~~of~~ in pose estimation. The ~~smoothing function averaged for every keypoint in position through successive frames. The~~ formula used to calculate and incorporate the landmark smoothing mechanism is

$$S_i = (1 - \alpha)P_i + \alpha C_i \tag{1}$$

where,

S_i = Smoothed coordinates (x,y) of landmark i after smoothing

P_i = (x,y) coordinates of the landmark in previous frame

C_i = (x,y) coordinates of the landmark in current frame

α = Smoothing factor

The smoothing factor reduces jitter in detected keypoints across frames by averaging the current frame's coordinates with previous frames. It does not ~~affect which detections are kept, but changes how detected~~ change the confidence-based acceptance rule, but it does alter how accepted keypoints are positioned and stabilized over time. A smoothing factor of 0.5 was chosen to achieve a balanced trade-off between responsiveness and stability in keypoint tracking. Lower values made the output overly sluggish, which caused delays in reflecting quick posture changes. Higher values introduced instability by allowing too much jitter from rapid movements. A value of 0.5 provided sufficiently smooth trajectories without sacrificing the system's ability to capture real-time posture dynamics accurately.

3.3 ~~Ergonomically Risky~~ Multi-frame Tracking and Activity ~~Detection~~Segmentation

~~This module classifies worker activities using pose landmarks and evaluates ergonomic risk according to the REBA method [?]. REBA provides standardized scoring to reflect the severity of bending and other postural loads. In this study, we focused on bending , with the trunk angle (i.e., forward flexion of the torso) serving as a key parameter in REBA scoring. The focus of this study was solely on trunk flexion because the primary objective was to determine ergonomic risk based on bending posture statistics of workers~~Because ergonomic risk assessment in video requires temporal continuity, detections belonging to the same worker must be associated across frames. We therefore use a centroid-based tracker in which each detected person bounding box is represented by its centroid, and each centroid in the current frame is matched to the nearest centroid from the previous frame within a fixed pixel-distance threshold. This lightweight association step provides persistent person IDs for posture-duration measurement and bending-frequency tracking while remaining suitable for real-time processing.

The tracker also defines how activity segments begin and end. When a worker remains matched across consecutive frames, the same ID is preserved and the standing or bending state is updated continuously. When a worker leaves the field of view and no valid match is found, the active segment is closed at the last valid frame. If that worker later reappears, the new detection is treated as a new tracking segment rather than a continuation of the previous one. This behavior avoids assigning unsupported durations through periods of non-visibility, but it also means that posture duration can be underestimated when workers move out of frame, which is a limitation of the current single-camera implementation.

3.4 Ergonomically Risky Activity Detection

This module classifies worker activities using pose landmarks and evaluates ergonomic risk ~~according to the REBA method [?]. REBA provides standardized scoring to reflect the severity of bending and other postural loads. In this study, we focused on bending, with the trunk angle (i.e., forward flexion of the torso) serving as a key parameter in REBA scoring. The focus of this study was solely on trunk flexion because the primary objective was to determine ergonomic risk based on bending posture statistics of workers. Since bending frequency and duration are central to trunk flexion assessment, this study applies REBA scoring only to the trunk flexion component. The present implementation estimates trunk flexion from~~ using the full REBA framework [?]. The system first detects body keypoints using YOLO pose and then computes posture angles for the neck, trunk, upper arms, lower arms, wrists, and legs from smoothed 2D ~~image coordinates rather than reconstructed 3D joint geometry. This choice supports low-cost deployment with a single RGB camera, but it also introduces viewpoint dependence because apparent angles in the image plane can change when the worker rotates relative to~~ keypoint coordinates. These segment angles are converted into REBA segment scores, combined through REBA Table A, Table B, and Table C, adjusted with a load/force score, and finally mapped to the corresponding REBA risk level. In this way, the ~~camera. Prior work comparing 2D and 3D camera-based joint-angle estimation has shown that camera configuration and dimensionality affect angle accuracy, which supports treating the present 2D formulation as an approximation rather than a full 3D biomechanical measurement [?]~~reported metric is a full REBA score rather than a trunk-only proxy.

Joint angles are derived from vectors formed by adjacent body landmarks. For any two vectors $\vec{u}$ and $\vec{v}$, the included angle is calculated using the dot-product relation

$$\theta = \arccos\left(\frac{\vec{u}\cdot\vec{v}}{\|\vec{u}\|\|\vec{v}\|}\right) \times \frac{180}{\pi}. \tag{2}$$

Trunk flexion is computed from the hip-midpoint to shoulder-midpoint vector against the vertical reference; neck flexion is estimated from the shoulder-midpoint to head vector against the trunk vector; upper-arm angle is measured between the upper arm and trunk; lower-arm angle is measured between the upper-arm and forearm vectors; wrist posture is estimated from the forearm-to-hand configuration; and leg posture is determined from knee flexion and lower-limb support geometry.

The resulting angles are converted into REBA segment scores using the standard rule structure: trunk posture is scored from neutral to severe flexion, neck posture is scored from neutral to flexed

posture, upper-arm posture is scored across increasing elevation ranges, lower-arm posture is scored according to deviation from the mid-range elbow position, wrist posture is scored by wrist deviation, and the legs are scored from stable bilateral support to bent or kneeling postures. Trunk, neck, and leg scores are combined through REBA Table A to obtain Score A. Upper-arm, lower-arm, and wrist scores are combined through REBA Table B to obtain Score B. Score A and Score B are then merged through REBA Table C to produce the final REBA score.

For handled tools or materials visible in the frame, a vision-language model is used to identify the object category and support approximate load/force estimation based on typical weight ranges of common construction tools and materials. This estimated load score is then incorporated into the REBA scoring workflow. Coupling is treated as a neutral default when it cannot be inferred reliably from the image. ~~Since bending frequency and duration are central to trunk flexion assessment, this study applies REBA scoring only to the trunk flexion component.~~

Let $\vec{HS}$ be the vector from the hip midpoint (H_x, H_y) to the shoulder midpoint (S_x, S_y), and let $\vec{V}$ be the vertical reference vector. We define:

$$\vec{HS} = (S_x - H_x,\ S_y - H_y) \tag{3}$$

$$\vec{V} = (0,\ 1) \tag{4}$$

The trunk angle θ_{trunk} is computed as follows:

$$\theta_{\text{trunk}} = \arccos\Big(\frac{\vec{HS} \cdot \vec{V}}{\|\vec{HS}\|\|\vec{V}\|}\Big) \times \frac{180}{\pi}, \tag{5}$$

where θ_{trunk} is measured in degrees. ~~According to REBA, the trunk flexion score for bending is typically assigned as shown in Table ??. Note that this table covers only the trunk score. This method can be used to compute the full REBA assessment as part of future work. The final REBA score also considers other body segments (e.g., neck, legs, arms), coupling (e.g., load handling), and any twisting or lateral bending. For example, if the trunk is also twisted or laterally bent, additional points are added to the trunk score. According to REBA, the trunk flexion score for bending is typically assigned as shown in Table ??. Note that this table covers only the trunk score. This method can be used to compute the full REBA assessment as part of future work. The final REBA score also considers other body segments (e.g., neck, legs, arms), coupling (e.g., load handling), and any twisting or lateral bending. For example, if the trunk is also twisted or laterally bent, additional points are added to the trunk score~~Table 1 summarizes the segment-level REBA rules implemented in the revised system.

Table 1: Implemented REBA segment scoring rules

Segment	Rule	REBA Score
Trunk	$< 5°$, $5°$–$20°$, $20°$–$60°$, $> 60°$	1, 2, 3, 4
Neck	$\leq 20°$, $> 20°$	1, 2
Upper arm	$< 20°$, $20°$–$45°$, $45°$–$90°$, $> 90°$	1, 2, 3, 4
Lower arm	$60°$–$100°$, otherwise	1, 2
Wrist	within $\pm 15°$, otherwise	1, 2
Legs	stable support, bent/unstable, kneeling/deep flexion	1, 2–4

In this research, a posture is classified as bending when the trunk angle θ_{trunk} exceeds a specified lower threshold (e.g., 20°). Depending on the measured trunk angle, ~~a REBA trunk score of 2, 3, or 4 is assigned (see Table ??)a REBA trunk score of 2, 3, or 4 is assigned (see Table ??)~~the corresponding segment contribution is determined using Table 1. A threshold of 20° was selected for the bending/standing classification based on the REBA framework [**?**]. The system also tracks how many times a worker transitions into a bending posture. Whenever the posture changes from standing to bending, it increments the bending count n_b:

$$n_b = \sum_{i=1}^{n} 1\Big(S_{\text{prev},i} = \text{Standing}\cancel{-} \text{ and } S_{\text{current},i} = \text{Bending}\Big), \tag{6}$$

where,

n_b = Total number of bending transitions;

$S_{\text{prev},i}$ = Previous posture state;

$S_{\text{current},i}$ = Current posture state;

$1(\cdot)$ = Indicator function;

n = Total number of time steps.

A timer tracks the time spent in each bending episode. For n identified bending periods, the total duration D_a is

$$D_a = \sum_{i=1}^{n_b} \big(T_{\text{end},i} - T_{\text{start},i}\big) \tag{7}$$

where,

D_a = Total bending duration;

$T_{\text{end},i}$ = End time of the ith bend;

$T_{\text{start},i}$ = Start time of the ith bend;

n_b = Total number of bending transitions.

In real time, the system ~~measures the trunk angle, calculates the REBA trunk score, measures the trunk angle, calculates the REBA trunk score~~computes segment angles, converts them to REBA segment scores, performs the Table A/B/C combination, adds the VLM-assisted load/force estimate, and updates bending duration and count. These ~~factors factors~~ outputs are used to ~~estimate an overall REBA score or to trigger alerts if bending duration and count exceed recommended thresholds. When a worker is flagged for prolonged bending or frequent transitions into bending, further ergonomic analysis may be required to determine the overall REBA risk level and implement risk mitigation strategies. estimate an overall REBA scoreor to trigger alerts if bending duration and count exceed recommended thresholds. In addition, when a worker is handling a visible tool or object, a vision-language model can be used to identify that item and map it to an approximate REBA load category based on typical weight ranges associated with common construction tools and materials. This provides a practical knowledge-guided estimate for the REBA load/force component when direct measurement is unavailable. When a worker is flagged for prolonged bending or frequent transitions into bending, further ergonomic analysis may be required to determine the overall REBA risk leveland implement risk mitigation strategies~~determine the final REBA score, assign the corresponding risk level, and trigger alerts when ergonomic thresholds are exceeded.

3.5 Risk Mitigation Recommendations Using VLM and RAG

This module generates individualized ergonomic risk recommendations through a ~~two stage~~ two-stage pipeline that uses both a Vision Language Model (VLM) and a Large Language Model (LLM). Figure 4 shows the overall framework. First, the system assembles the inputs needed for recommendation. A VLM such as Qwen2 or Molmo [**?**] produces a text description of the construction activity from an input image, including salient actions, tools, and observable context. An example VLM prompt is provided in Table 2. Posture statistics are obtained from the YOLOv8 method described earlier, including bending counts and bending duration for each individual. These elements are programmatically inserted into a template, where placeholders enclosed in angle brackets represent content returned by the models.

Second, the system generates recommendations with ~~retrieval augmented~~ retrieval-augmented generation (RAG). ~~Relevant standards and public sources (e.g., [**?**]), including safety guidelines and incident statistics, form the knowledge base. The RAG layer retrieves passages that match the current activity description and posture statistics. An LLM accessed through public APIs~~

~~(for example ChatGPT, DeepSeek, or LLaMA)~~ The implementation uses ChromaDB as the vector database, the *all-MiniLM-L6-v2* model to embed text chunks, *gpt-4-turbo* as the language model, and Streamlit as the user interface. The knowledge base consists of 126 embedded text chunks derived from *Simple Solutions: Ergonomics for Construction Workers* [**?**]. Retrieval is performed with normalized cosine similarity, and only passages above a relevance threshold are forwarded to the generator. The LLM then composes answers to the three targets specified in Table 2: causes of risk, mitigation strategies, and corrective actions. The recommendations are grounded by the retrieved passages and are returned with ~~the~~ supporting snippets for transparency. ~~To ensure usefulness and accuracy, the recommendations produced by the VLM–RAG–LLM pipeline are reviewed by domain experts in construction safety and ergonomics, and the retrieved content is cross-checked against authoritative safety guidelines before finalization. To ensure usefulness and accuracy, the recommendations produced by the VLM–RAG–LLM pipeline are reviewed by domain experts in construction safety and ergonomics, and the retrieved content is cross-checked against authoritative safety guidelines before finalization.~~ In the present study, however, the quality and effectiveness of the generated recommendations were not formally evaluated through expert scoring, inter-rater agreement, user studies, or task-level outcome analysis.

Accordingly, the VLM–RAG component should be interpreted as a proof-of-concept recommendation module whose outputs are grounded in retrieved ergonomic guidance, but whose practical effectiveness has not yet been systematically validated. A formal evaluation of recommendation quality remains an important direction for future work.

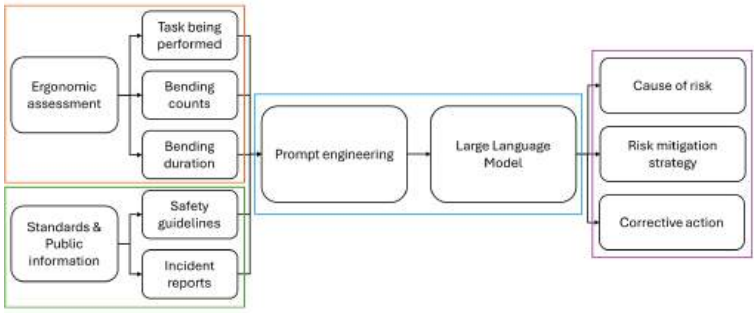


Figure 4: An information flow of generating recommendations for risk mitigation strategy using LLMs

Table 2: Prompts used for activity description and risk analysis

Prompt	Description
VLM Prompt	`Please provide a detailed description of the construction worker's activity in this image, including specific actions, tools being used, and the work context if identifiable.`

LLM Prompt	`Based on the following description of a construction worker's activity from an image: <<Task description>>. <<Posture related statistics>>.`
	
	`Please analyze the worker's activity and provide:`
	`1. <<Causes of risk>>`
	`2. <<Risk mitigation strategies>>`
	`3. <<Corrective actions>>`

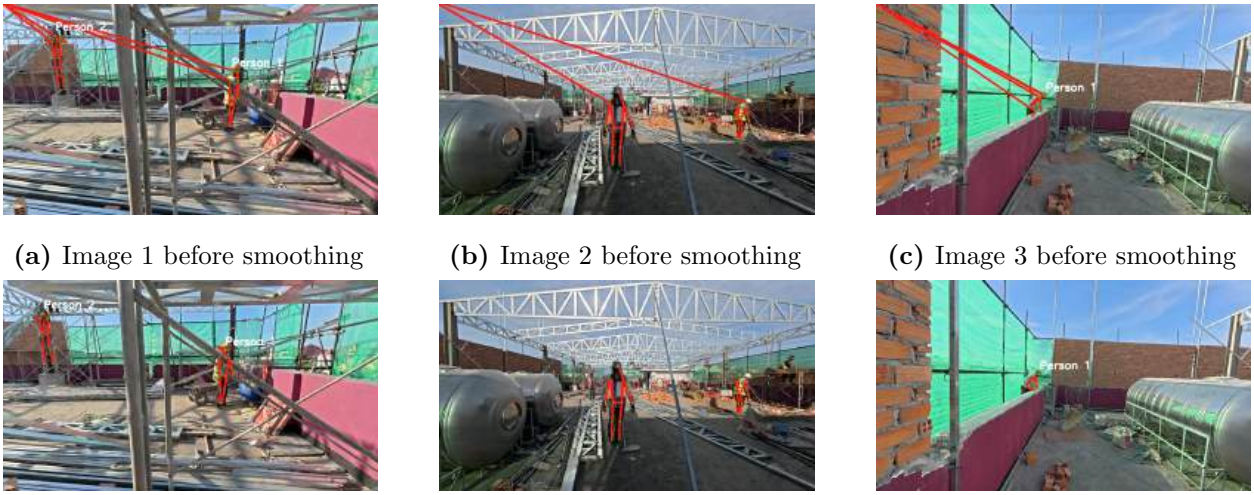
4 Case Study

In this research, the system was initially tested using a regular webcam. The experiment was conducted on an Apple MacBook Pro with M3 chip. The system had 18 GB memory and ran on macOS Sequoia 15.1.1. 20 Creative Commons–licensed videos were gathered. They covered five construction activities: drywall finishing, tying rebar, kneeling creepers, hand screeding, and laying concrete block, with four videos for each activity. The videos range from 10 to 20 minutes in duration and feature an average of 2–3 participants per video, which resulted in approximately 40-60 participants in total. All recordings were captured in real construction site environments, typically filmed by a moving person. Camera heights generally ranged from approximately 1.2 to 1.7 meters above the ground that corresponded to chest or eye level of the videographer. For analysis, each video was processed into frames at a rate of one frame per second. During pose detection, every second frame was skipped to reduce computational load while preserving critical motion details. The accuracy of pose detection and activity classification was manually verified by the authors to ensure the system effectively flagged critical ergonomic risks.

4.1 Landmark Smoothing

The landmark smoothing technique is used to enhance the system's ability to generate more refined and accurate representations of workers' postures during continuous movement. Figure 5 illustrates the impact of landmark smoothing on pose estimation using YOLOv8. Figures 5a, 5b, and 5c show keypoint positions without smoothing~~, where minor fluctuations between frames create inconsistencies. These variations can lead to inaccuracies in dynamic pose analysis, particularly in real-time applications.~~ . In these unsmoothed examples, some low-confidence keypoints are

returned by the pose model as the default image coordinate $(0, 0)$, which is the upper-left corner of the frame. Therefore, the problem shown in Figure 5 is not that the worker's landmark truly belongs at $(0, 0)$, but that a missing or unreliable detection is encoded by the model output at that origin coordinate. If these raw coordinates are used directly, the skeleton suddenly stretches toward the upper-left corner and creates a physically implausible posture. In contrast, Figures 5d, 5e, and 5f present the results after applying landmark smoothing, which revealed more stable and consistent keypoint trajectories. The smoothing algorithm reduces abrupt changes by averaging each frame's landmark positions with those from previous frames by using a specified smoothing factor. As a result, the method functions primarily as a missing-data interpolation mechanism in these examples, because it prevents a single spurious $(0, 0)$ output from dominating the skeleton, while also reducing ordinary frame-to-frame jitter. This process stabilizes the skeletal representation, which ensures a more accurate depiction of movement over time.



(a) Image 1 before smoothing **(b)** Image 2 before smoothing **(c)** Image 3 before smoothing

(d) Image 1 after smoothing **(e)** Image 2 after smoothing **(f)** Image 3 after smoothing

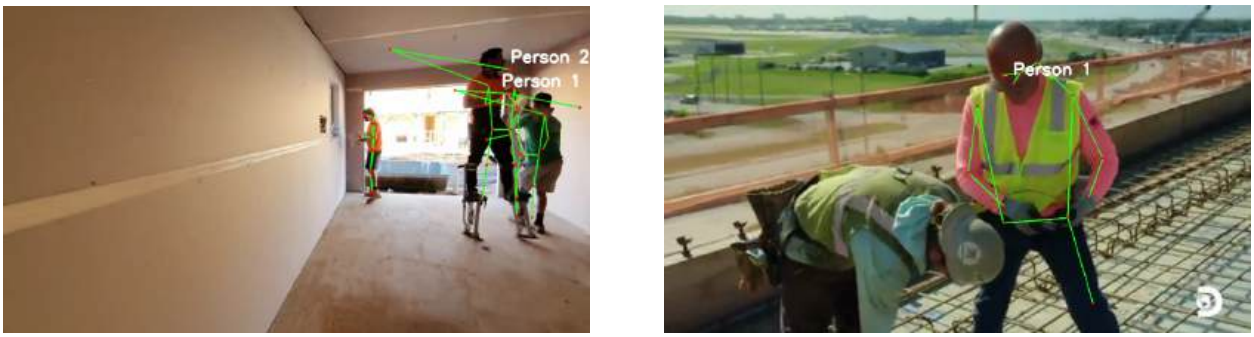
Figure 5: Comparison of pose estimation with and without landmark smoothing

4.2 Failure Scenarios Under Occlusion and Multi-person Overlap

Although the qualitative examples in Figure 7 demonstrate successful posture extraction, the model does not perform equally well in all site conditions. Figure 6 presents two representative failure scenarios observed in the construction videos. In Figure 6a, multiple workers appear at different depths in a narrow indoor workspace. The distant worker is only partially visible, and the workers on the right overlap spatially, which causes elongated or misplaced skeleton connections. In Figure 6b, one worker bends forward while another stands directly beside him, which creates partial body

occlusion and ambiguity in separating the two individuals. In both cases, the output remains qualitatively informative, but individual joints can be misplaced and some body segments may be omitted or incorrectly connected.

These cases highlight an important limitation of the current single-camera YOLOv8-based pipeline. Performance degrades when workers overlap, when limbs are self-occluded, or when a person occupies only a small portion of the frame. Consequently, downstream measurements such as bending duration, transition counts, and REBA-based posture interpretation may become less reliable in crowded or heavily occluded scenes. Future improvements may include stronger multi-object tracking, temporal keypoint interpolation across longer windows, and multi-camera views to reduce visibility loss.



(a) Pose estimation under multi-person overlap and depth variation

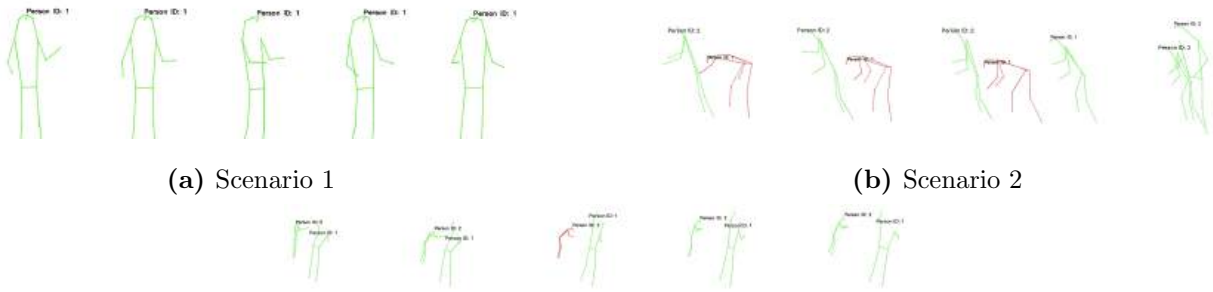
(b) Pose estimation under occlusion during a bending task

Figure 6: Failure scenarios for skeleton extraction in crowded and partially occluded construction scenes

4.3 Detection of Activity with Ergonomic Risks

Figure 7 shows skeletal overlays of workers performing drywall finishing tasks in three different scenarios. Drywall finishing involves the process of smoothing, taping, and sealing joints between drywall sheets after they are mounted on walls or ceilings. This work often requires individuals to bend and crouch, as they need to reach lower sections of the wall or work in confined spaces to apply compound and tape properly. Prolonged bending during drywall finishing can strain the lower back, which increases the risk of injury. Figure 7a shows a single worker labeled as ”“Person 1,”” who remains standing across multiple frames. The skeletal overlays display straight lines that represent the worker's upright posture. It shows that the skeletons remain stable throughout the sequence. The figure also shows that no red lines appear in these frames, which means that no bending was detected. Figure 7b includes two workers labeled as ”“Person 1~~” and ”~~” and “Person 2.”” Each

worker seem to perform different activities over time. ~~"~~"Person 1~~"~~" moves between normal and risky postures. ~~"~~"Person 2~~"~~" remains in normal posture without changes. Figure 7c illustrates the activities of ~~"~~"Person 2~~"~~" across a sequence of frames. The worker alternates between standing and bending.

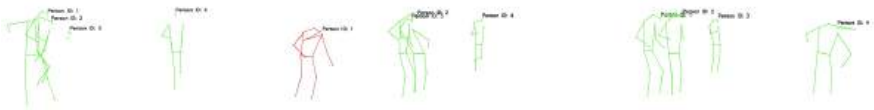


(a) Scenario 1 **(b)** Scenario 2

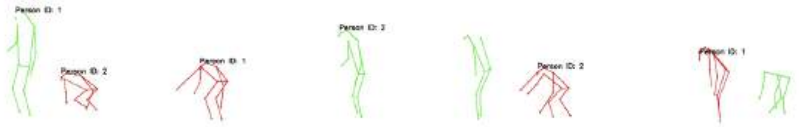
(c) Scenario 3

Figure 7: Skeleton Representation of Workers Performing Drywall Finishing (Red: Ergonomic Risk Activity)

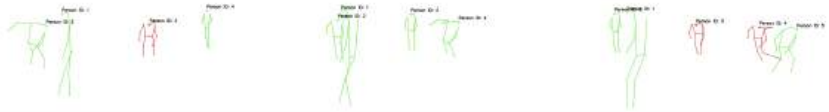
Figure 8 depicts skeletal overlays of workers' performing hand screeding. Hand screeding is a labor intensive process used in construction to level and smooth concrete surfaces. This activity uses a flat board or screed to spread the concrete evenly across the surface. Workers often need to bend and stoop while performing this task to ensure an even and level finish. During hand screeding, repetitive bending and awkward postures can place significant strain on the back, especially during prolonged tasks. Figure 8a shows multiple workers labeled as ~~"~~"Person 1~~" to "~~" to "Person 4.~~"~~" The majority are in normal posture, represented by green skeletons, while ~~"~~"Person 1~~"~~" transitions into bending, shown with red skeletons. Figure 8b focuses on two workers, ~~"~~"Person 1~~" and "~~" and "Person 2,~~"~~" actively alternating between standing and bending. ~~"~~"Person 1~~"~~" shows a mix of standing and bending postures, visualized through alternating green and red skeletons. ~~"~~"Person 2~~"~~" primarily shows ergonomically risky postures, represented by red skeletons. Figure 8c includes a group of workers labeled from ~~"~~"Person 1~~" to "~~" to "Person 6.~~"~~" The skeletal overlays illustrate a variety of activities, with some individuals standing consistently and others bending intermittently. ~~"~~"Person 3~~" and "~~" and "Person 5~~"~~" exhibit frequent bending motions, indicated by red skeletons, while ~~"~~"Person 4~~" and "~~" and "Person 6~~"~~" remain standing throughout the sequence.



(a) Scenario 1



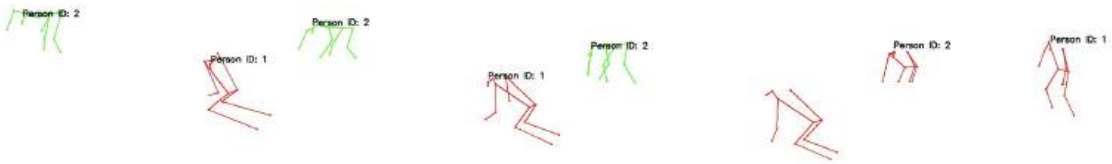
(b) Scenario 2



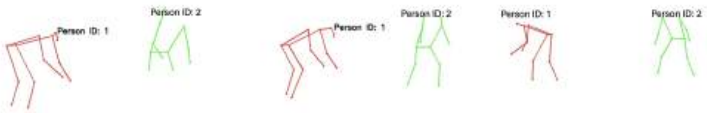
(c) Scenario 3

Figure 8: Skeleton representation of workers performing hand screeding over consecutive frames at 1-second intervals

Figure 9 depicts skeletal overlays of workers' performing kneeling finishing. Kneeling finishing are used in construction and other manual tasks where workers need to spend extended periods kneeling on hard surfaces, such as during tasks like tiling, plumbing, or electrical work. Kneeling for long durations can lead to discomfort, joint pain, and strain on the knees and lower back. This repetitive pressure can be physically taxing, especially in environments where the worker must stay in a kneeling position for long periods. Figure 9a shows two workers labeled as ”“Person 1”~~ and~~ ”” and “Person 2.”~~ ”~~” “Person 1”~~ ”~~ is predominantly engaged in bending postures, represented by red skeletons, while ”“Person 2”~~ ”~~ alternates between standing and bending. Figure 9b focuses on ”“Person 1,”~~ ”~~ demonstrating repeated transitions between bending and standing postures. The red skeletons depict periods of kneeling, while the green skeletons represent standing intervals. Figure 9c includes a detailed sequence for ”“Person 1,”~~ ”~~ which shows transitions from bending to standing over several frames.



(a) Scenario 1

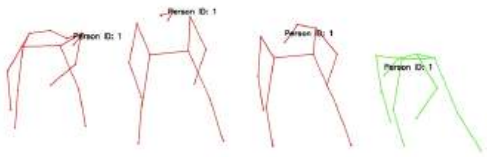


(b) Scenario 2

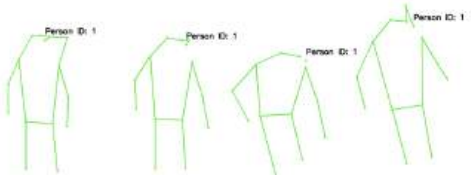
(c) Scenario 3

Figure 9: Skeleton representation of workers performing kneeling creepers over consecutive frames at 1-second intervals

Figure 10 depicts skeletal overlays of workers' laying concrete blocks. Laying concrete blocks is a physically demanding task that requires workers to lift, position, and align heavy blocks to create walls or other structures. This often involves bending, lifting, and repetitive motions, which can cause strain on the back, shoulders, and arms, especially when working with traditional, heavy concrete blocks. Constantly lifting and manipulating these blocks over extended periods can lead to fatigue and increase the risk of musculoskeletal injuries. Figure 10a shows ~~"~~ "Person 1~~"~~" engaged in bending activities across multiple frames. Figure 10c includes two workers, ~~"~~ "Person 1~~" and "~~" and "Person 2.~~" "~~" "Person 1~~"~~" alternates between standing and bending postures, shown by the mix of green and red skeletons, while ~~"~~ "Person 2~~"~~" remains consistently standing. Figure 10b focuses on ~~"~~ "Person 1~~"~~" maintaining a standing posture across several frames.



(a) Scenario 1

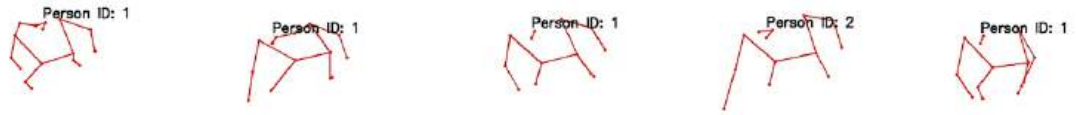


(b) Scenario 3

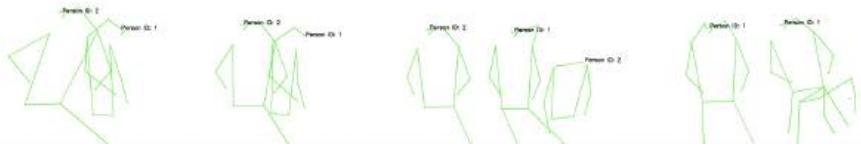
(c) Scenario 2

Figure 10: Skeleton representation of workers laying concrete blocks over consecutive frames at 1-second intervals

Figure 11 depicts skeletal overlays of workers' tying rebar. Tying rebar requires workers to bend, twist, and secure metal rebar with wire ties. This task involves repetitive hand movements and awkward postures, often requiring workers to bend down or crouch for long periods. This can lead to strain on the back, hands, and wrists. The repetitive nature of tying rebar also increases the risk of hand injuries or discomfort from prolonged gripping. Figure 11a shows two workers, "Person 1" ~~and "~~" and "Person 2," " predominantly engaged in bending activities. The red skeletons represent bending postures consistently, with multiple frames showing "Person 1" " bending, while "Person 2" " is seen bending in the middle of the sequence. Figure 11b features "Person 1" ~~and "~~" and "Person 2," " both transitioning between standing and bending postures. "Person 1" " shows a more upright posture in most frames, while "Person 2" " demonstrates intermittent bending. Figure 11c focuses on the activity of "Person 2" " and demonstrates transitions from standing to bending and back to standing.



(a) Scenario 1



(b) Scenario 2

(c) Scenario 3

Figure 11: Skeleton representation of workers tying rebar over consecutive frames at 1-second intervals

4.4 Activity Duration and Count

The developed system is also able to track the duration and count of normal and risky postures, as well as how often workers switch between them. Figure 12 illustrates an example to track and visualize activity duration and bending count for a construction worker labeled as ~~"~~"Person 1.~~"~~". The worker's posture is classified as ~~"Bending"~~ "Bending" (red) and ~~"Standing"~~ "Standing" (green) based on the alignment of key body landmarks such as the nose, shoulders, hips, and ankles. The bending duration is shown in red, with times of 5 seconds, 3 seconds, and 2 seconds recorded for three separate intervals. Similarly, the standing duration is shown in green, with times of 1 second and 4 seconds between bending intervals. The bending count is updated each time the worker transitions from standing to bending, as indicated by the arrows in the image. In this example, the bending count increases by 1 twice during the session, resulting in a total bending count of 2.

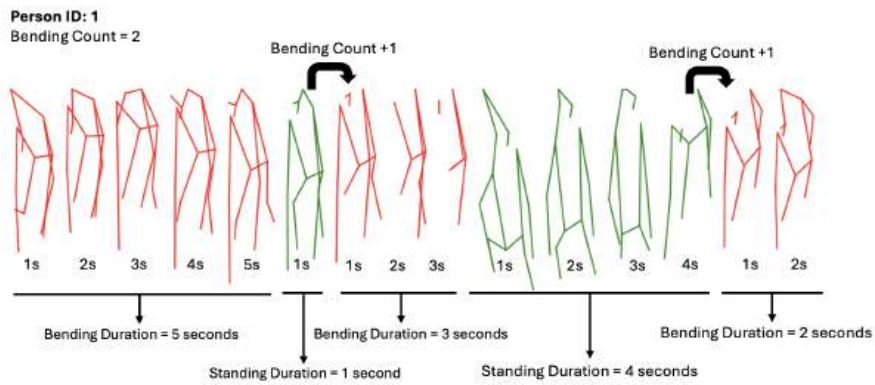


Figure 12: Displaying Activity Duration and Bending Count

4.5 Comparison of Model and Manual Evaluation

We conducted a comparison of model generated and manually recorded activity data to evaluate the system's accuracy in detecting ergonomic risk factors. To establish a ground truth for comparison, human observers manually reviewed recorded videos of workers performing these scenarios. They identified postural changes and recorded their duration and count. By comparing the model's detections with these manually recorded values, we calculated the error rate. The error rate quantifies the percentage difference between the two methods. The error rate was determined using the following formula:

$$E_{\text{final}} = \frac{100}{3}\left(\frac{|M_{\text{np}} - G_{\text{np}}|}{G_{\text{np}}} + \frac{|M_{\text{er}} - G_{\text{er}}|}{G_{\text{er}}} + \frac{|M_{\text{freq}} - G_{\text{freq}}|}{G_{\text{freq}}}\right) \tag{8}$$

where,

E_{final} = Final average error percentage;

M_{np}, G_{np} = Model and ground truth values for normal posture duration;

M_{er}, G_{er} = Model and ground truth values for ergonomically risky duration;

M_{freq}, G_{freq} = Model and ground truth values for count of risky activity.

Table 3 shows a detailed comparison of manual and model evaluations for different construction related tasks. Each task is broken down into multiple test scenarios. The table reports three main metrics for each scenario: duration of normal posture in seconds, duration of ergonomically risky posture in seconds, and the count of risky activities. The final column presents the error rate, which represents the difference between the model's results and manual observations. Many scenarios show low error rates. This indicates that the model tracks posture durations and frequencies accurately. For instance, ~~"~~"Hand Screeding Scenario 3~~" and "~~" and "Kneeling Creeper Scenario 2~~"~~" both

achieve a 0 percent error rate. In these tests, the model's metrics match the manual measurements perfectly. Tasks like ~~"~~"Hand Screeding Scenario 1~~"~~" also demonstrate a fairly low error rate of 2.45 percent. Some scenarios reveal moderate to higher discrepancies. ~~"~~"Tying Rebar Scenario 1~~"~~" shows an error rate of 5.36 percent. This higher value suggests that the model may struggle with frequent or subtle changes in posture. Rapid transitions between normal and risky positions can create more counting errors. Minor variations in how workers perform each task may further complicate detection. Sensor calibration or environmental factors like lighting can also affect the model's accuracy. Activities like ~~"~~"Hand Screeding Scenario 2~~" and "~~" and "Kneeling Creeper Scenario 3~~"~~" display error rates of 5.13 percent and 6.06 percent, respectively. These rates highlight potential challenges in distinguishing borderline postures or counting movements that happen quickly. Consistent repetition or variations in a task can lead to partial misclassifications. As a result, the model's readings may differ from the manual ground truth. In contrast, ~~"~~"Laying Concrete Block Scenario 1~~" and "~~" and "Tying Rebar Scenario 2~~"~~" each show an error rate of 0 percent. These scenarios point to conditions where the model excels at identifying posture transitions. Fewer rapid movements or clearer distinctions between normal and risky postures can reduce miscalculations. Stable lighting or simpler task patterns may also support accurate posture detection.

Table 3: Comparison of ~~Model~~ model and ~~Manual Evaluations~~manual bending estimates

~~Activity~~Task	Case	Model Bend Dur.	Manual Bend Dur.	Model Bend Cnt.	Manual Bend Cnt.	Error (%)
	1	12	12	92	90	2.45
Hand Screeding	2	22	22	21	100	5.13
	3	7	7	47	47	0.00
	1	10	10	9	3	3.13
Drywall Finish	2	7	7	33	33	2.02
	3	3	3	15	15	0.00
	1	0	0	31	0	0.00
Concrete Block	2	5	5	19	19	2.50
	3	13	13	78	78	4.79
	1	9	9	25	25	0.00
Kneeling Creeper	2	5	5	0	0	0.00
	3	6	6	8	14	6.06
	1	12	12	32	30	5.36
Tying Rebar	2	4	4	2	4	0.00
	3	9	9	31	31	1.33

4.6 Full-body REBA Scoring Examples

To demonstrate that the revised system computes full-body REBA rather than only trunk-based scoring, five representative examples are shown in Figures 13 and 14. In each case, the system detects body keypoints, computes segment angles, converts them into segment-level REBA scores, combines them through Table A, Table B, and Table C, adds the load/force score, and outputs the final REBA score and associated risk level. The load/force term in these examples is obtained through the VLM-assisted approximate estimation workflow described in Section 3.4.

Cases 1, 4, and 5 produce high or very high REBA scores because they combine pronounced trunk flexion with elevated arm posture and unstable lower-limb support. Cases 2 and 3 produce medium-risk scores, reflecting less severe whole-body loading despite visible non-neutral posture. These examples illustrate that the revised pipeline now evaluates neck, trunk, upper arm, lower arm, wrist, legs, and load/force together, rather than reporting only a trunk-derived index.

Table 4: Summary of five full-body REBA scoring examples

Case	Score A	Score B	Load	Final REBA	Risk Level
Case 1	7	5	2	9	High
Case 2	5	4	2	5	Medium
Case 3	4	4	2	4	Medium
Case 4	10	4	2	11	Very High
Case 5	8	4	2	9	High

(a) Case 1: final REBA score 9 (High)

(b) Case 2: final REBA score 5 (Medium)

(c) Case 3: final REBA score 4 (Medium)

(d) Case 4: final REBA score 11 (Very High)

Figure 13: Full-body REBA scoring examples for Cases 1–4. Each panel shows the original video frame and the corresponding segment-level REBA interpretation.

(a) Case 5: final REBA score 9 (High)

Figure 14: Additional full-body REBA scoring example for Case 5.

4.7 Benchmark Against Existing Pose Estimation Pipelines

To provide a direct quantitative comparison with an established baseline, we benchmarked three pose-estimation configurations on five representative clips from the construction dataset: OpenPose, YOLOv8 raw, and YOLOv8 with landmark smoothing. Each clip contained 10 sampled frames, and all methods were evaluated using the same downstream processing pipeline for keypoint filtering, ~~trunk-angle~~ segment-angle estimation, bending detection, and full-body REBA scoring.

The benchmark reports required-keypoint visibility, required-joint coverage, mean and maximum REBA score, REBA standard deviation, mean trunk-angle jitter, proportion of bending frames, and runtime per frame.

Table ?? summarizes the aggregate results. YOLOv8 with smoothing achieved the strongest overall balance between pose completeness and temporal stability, with mean required-joint coverage of 71.75%, mean trunk jitter of 8.69°, and runtime of 60.7 ms/frame. In contrast, OpenPose achieved only 46.16% mean required-joint coverage and 4.0% required-keypoint visibility, while requiring 2827.3 ms/frame. Compared with YOLOv8 raw, the smoothed YOLOv8 variant reduced trunk-angle jitter from 13.24° to 8.69° while preserving the same keypoint visibility and coverage statistics. These results support the choice of YOLOv8 for the present system and show that the smoothing step improves temporal stability of REBA-related angle estimates without sacrificing throughput.

~~Figures ?? and ?? present~~ Figure ?? presents all five benchmarked cases in a compact 3-column layout: OpenPose, YOLOv8 raw, and YOLOv8 with smoothing. Case 1 (drywall clip) shows that both YOLOv8 variants preserve more complete body structure than OpenPose, and smoothing reduces trunk jitter from 15.31° to 13.13° while increasing the mean REBA score from 4.86 to 5.43 by preserving a more pronounced bending posture. Case 2 (hand1 clip) highlights a stronger temporal stabilization effect, where smoothing lowers trunk jitter from 18.32° to 10.08° while keeping the same bending-frame proportion of 60%; OpenPose detects substantially fewer bending frames at 30%. Case 3 (kneeling clip) shows the clearest jitter reduction, from 4.08° to 0.13°, indicating that smoothing is especially beneficial when posture changes are subtle and the worker remains relatively compact in the image. Case 4 (block clip) shows complete required-keypoint visibility and joint coverage for both YOLOv8 variants, while smoothing lowers trunk jitter from 15.27° to 11.43° and produces a less extreme maximum REBA score than YOLOv8 raw. Case 5 (hand3 clip) is the most difficult of the five examples, with lower coverage for every method; even in this more challenging case, YOLOv8 variants substantially exceed OpenPose in joint coverage. We did not benchmark RTMPose, MMPose, VideoPose3D, or Poserac in the present study, so we do not claim superiority over those methods. Instead, we identify them as important future baselines for broader evaluation, especially for temporal pose refinement and repetitive-action modeling.

Table 5: Aggregate benchmark comparison across five construction clips

Method	~~Vis. (%)~~Cov. (%)	Mean REBA	~~Max REBA / REBA SD~~Jitter (°)	Bend (%)	Runtime (ms/frame)
OpenPose	~~4.00~~46.16	4.58	~~6.0 / 0.88~~20.21	32.0	2827.3
YOLOv8 raw	~~30.79~~71.75	4.67	~~7.6 / 1.64~~13.24	44.0	62.05
YOLOv8 + smoothing	~~30.79~~71.75	4.78	~~7.4 / 1.60~~8.69	44.0	60.7

~~Hand 1 / OpenPose Hand 1 / YOLOv8 raw Hand 1 / YOLOv8 + smoothing~~

~~Kneeling / OpenPose Kneeling / YOLOv8 raw Kneeling / YOLOv8 + smoothing~~

~~Benchmark cases 1–3 arranged as 3 rows by 3 methods. Each row corresponds to one clip and compares OpenPose, YOLOv8 raw, and YOLOv8 with smoothing under the same downstream REBA workflow.~~

~~Block / OpenPose Block / YOLOv8 raw Block / YOLOv8 + smoothing~~

~~Hand 3 / OpenPose Hand 3 / YOLOv8 raw Hand 3 / YOLOv8 + smoothing Benchmark cases 4–5 arranged as 2 rows by 3 methods.~~

4.8 Recommendations based on VLM-RAG

Figure **??** shows the image of a worker tying rebar at a construction site, along with the prompt and the response generated by Molmo, a vision-language model [**?**]. The description emphasizes that the worker is adjusting, connecting, and securing the thin metal rods (tying rebar). This detailed account aids the system in understanding the work being performed and serves as an essential input for further ergonomic risk analysis and mitigation strategies. The response from RAG explains that workers can face risks such as poor posture, repeated bending, and awkward movements, all of which can lead to injuries. It offers advice on reducing these risks, such as using better tools and alternating tasks regularly.

In the current study, these example recommendations are presented to illustrate the behavior of the VLM–RAG pipeline rather than to claim validated recommendation quality. A systematic evaluation of whether the generated advice is consistently relevant, grounded, and practically effective has not yet been conducted.

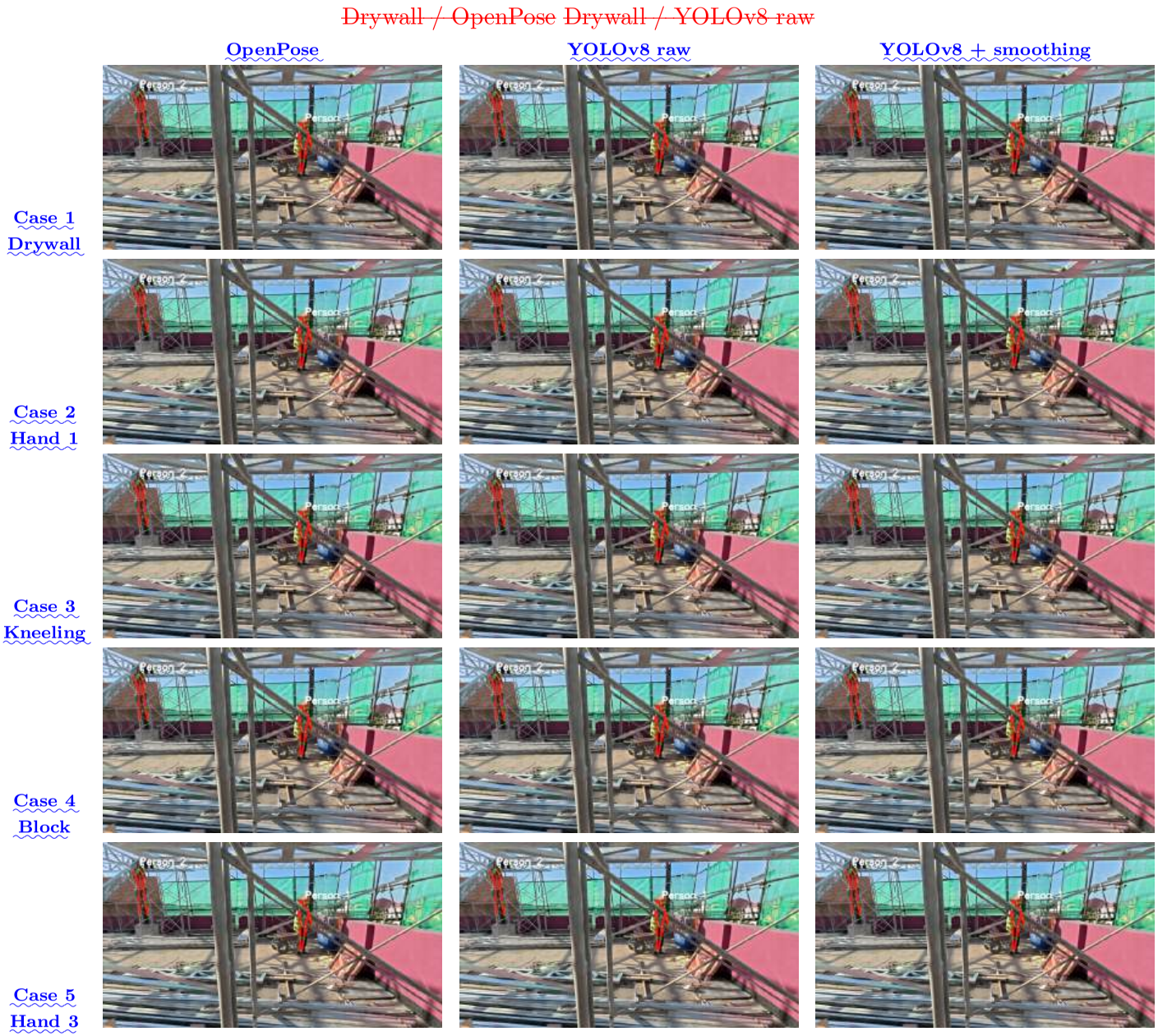


Figure 15: ~~Drywall / YOLOv8 + smoothing~~Compact five-case benchmark comparison shown on a single page. Each row corresponds to one clip and each column corresponds to one pose-estimation pipeline evaluated under the same downstream full-body REBA workflow.

Prompt: Please provide a detailed description of the construction worker's activity in this image, including specific actions, tools being used, and the work context if identifiable.

Response from Molmo: The image shows a construction worker engaged in a specific task on a construction site. The worker is kneeling on what appears to be a concrete surface, surrounded by a grid-like pattern of metal bars. This suggests they are working on rebar installation, which is a common practice in concrete construction to provide reinforcement. The worker is wearing protective gear, including a yellow hard hat and a yellow safety vest, which are standard safety requirements on construction sites. This indicates that the image is capturing a moment during active construction work. The most notable aspect of the image is the overlay of green lines and red dots on the worker's body. These markings are likely part of a motion capture or analysis system. The green lines appear to trace the worker's movements, while the red dots may indicate specific points of interest or joint articulation. This type of overlay is often used in biomechanical studies to analyze the ergonomics and efficiency of construction work. At the bottom of the image, there's a black bar with white text that provides specific information about the worker's activity. It states "Person 1 activity bending, activity time 0 seconds, bending frequency 3." This data suggests that the captured moment shows the worker performing a bending motion, possibly to pick up or manipulate the rebar. The brief activity time (0 seconds) and high frequency (3) indicate that this is likely a repetitive movement that the researchers are studying. The work context appears to be a concrete foundation or floor where rebar installation is taking place. The grid of metal bars surrounding the worker is typical of rebar placement patterns, and the concrete surface suggests that this could be part of a larger structure's foundation. Overall, this image seems to be part of a study or analysis of construction worker ergonomics, specifically focusing on the bending movements involved in rebar installation. The combination of the motion capture overlay, detailed text, and the visible work environment provides a comprehensive view of both the physical activity and the construction setting.

Figure 16: Response generated by Molmo

5 Conclusion

The proposed real-time ergonomic risk detection system is able to ~~effectively identifies and assesses~~ ~~effectively identifies and assesses~~ identify and assess construction workers' postures from ordinary video. The results demonstrate the system's capability to track bending and standing postures, measure activity duration, ~~and count transitions with a high degree of accuracy. The landmark smoothing techniques is able to improve stability in detecting workers' movements and reduce fluctuations due to rapid postural changes. and count transitions with a high degree of accuracy. The landmark smoothing techniques is able to improve stability in detecting workers' movements and reduce fluctuations due to rapid postural changes~~count posture transitions, and compute a full-body REBA score from YOLO pose keypoints. The revised workflow evaluates neck, trunk, upper-limb, wrist, and leg postures, combines them through REBA Tables A, B, and C, and incorporates a VLM-assisted approximate load/force estimate to assign the final risk level. Landmark smoothing improves temporal stability of the detected keypoints, and the benchmark shows that the YOLOv8-based pipeline provides a much stronger balance of speed and pose completeness than OpenPose on the five representative construction clips. By providing real-time feedback, the system enables safety managers to proactively identify and mitigate

ergonomic risks. ~~Practical applications include the development of automated alerts for workers engaging in prolonged bending, implementation of posture training programs, and integration with wearable sensors to enhance risk assessment. The validation against manual evaluations shows that the system's error rate is generally low, which confirms its reliability in identifying ergonomic risks. Practical applications include the development of automated alerts for workers engaging in prolonged bending, implementation of posture training programs, and integration with wearable sensors to enhance risk assessment. The validation against manual evaluations shows that the system's error rate is generally low, which confirms its reliability in identifying ergonomic risks.~~ A notable advantage of the proposed system is its cost-effectiveness and accessibility. Unlike traditional ergonomic assessment methods that often require expensive specialized equipment, this system can be implemented using standard hardware, a regular ~~commercial grade camera commercial grade camera~~commercial-grade camera, and a computer with moderate processing capabilities.

Despite its effectiveness, the system has some limitations. The accuracy of pose detection may decrease in cases where workers' movements are partially occluded or when multiple workers overlap in the camera's field of view. ~~Moreover, while the model can track posture transitions, it does not currently factor in external loads or other biomechanical stressors that contribute to ergonomic risks.~~ In addition, posture duration may be underestimated when a worker moves out of frame because the current activity segment is closed at the last valid detection and any later reappearance is treated as a new segment. Another important limitation is that trunk and limb angles are computed from 2D projections of body keypoints obtained from a single camera. When the worker's torso is rotated relative to the camera plane, the projected 2D angle may deviate from the true 3D ~~trunk~~ flexion angle, which can in turn affect REBA-related interpretation. This limitation is consistent with prior evidence that joint-angle estimates derived from 2D and 3D camera measurements can differ meaningfully depending on viewpoint and measurement setup [**?**] . ~~Moreover, while the model can track posture transitions, it does not currently factor in external loads or other biomechanical stressors that contribute to ergonomic risks~~The load/force term is also estimated approximately from the object or tool category inferred by the VLM rather than from direct physical measurement, so the assigned load score should be interpreted as a practical approximation. A further limitation is that the VLM–RAG recommendation module has not yet been evaluated by domain experts and has not been systematically validated through user studies, inter-rater agreement analysis, or task-level outcome measures. As a result, the current manuscript cannot claim that the generated recommendations consistently improve worker behavior or reduce

ergonomic risk in practice. Future improvements could involve refining the detection algorithm to handle more complex postures~~and integrating additional risk factors, such as external forces exerted during lifting and carrying tasks~~, incorporating 3D pose reconstruction or multi-camera capture, ~~and integrating additional risk factors such as external forces exerted during lifting and carrying tasks~~validating the VLM-assisted load estimates against measured weights, and conducting formal evaluations of recommendation quality using groundedness, relevance, actionability, usefulness, and human-subject feedback metrics. Future research should also focus on enhancing the system's real-time feedback mechanisms by incorporating predictive analytics to identify workers at risk before injuries occur. Additionally, integrating multimodal data, such as wearable sensor inputs and physiological indicators, could improve the system's accuracy in assessing ergonomic strain.